

Tinnaphop Sonsang

111/1 Moo 9, Nong Kratum Sub-district, Thap Than District, Uthai Thani

Email: tinnaphop.moss@gmail.com | Mobile: 064-948-1740

LinkedIn: [Tinnaphop Sonsang](#) | GitHub: github.com/badgaitintinagain, github.com/badgaitintin

**Not yet conscripted for military service, but eligible to obtain a gender certification as an LGBTQ+ individual for submission next year.*

This has no impact on work performance or contract.

PROFESSIONAL SUMMARY

Computer Engineering and AI graduate with hands-on experience in Machine Learning, Data Science, and Computer Vision. I have worked on AI and ML projects involving structured and image data, from data preparation and feature engineering to model development, evaluation, and deployment. My experience includes Python, SQL, statistical analysis, and cloud-based ML environments, with a strong interest in building practical solutions and continuously improving them through experimentation.

EDUCATION

Walailak University

B.Eng. Computer Engineering and Artificial Intelligence

GPAX: 3.65 (First-class honors) | Graduation: 7 May 2026

Nakhon Si Thammarat

June 2022 – 2026

CERTIFICATIONS

EF SET English Certificate ([View certificate: cert.efset.org/en/gCMcLJ](https://cert.efset.org/en/gCMcLJ))

Awarded 3 Aug 2026

- Overall score: 56/100 — B2 Upper Intermediate (CEFR); Reading: C1 Advanced (64/100)

KEY SKILLS

- AI & Machine Learning:** Python, PyTorch, Machine Learning, Deep Learning, Neural Networks, Model Development, Experimentation, Model Evaluation and Performance Testing.
- Computer Vision & Image Processing:** OpenCV, YOLO, RT-DETR, Grounding DINO, Object Detection, Image Processing, OCR, Object Tracking, Keypoint Detection, Depth Estimation, and Data Augmentation.
- Data Processing & Engineering:** Pandas, NumPy, SQL, Data Gathering, Data Cleansing, Data Preprocessing, Feature Engineering, Data Quality Control, Dataset Deduplication, and Exploratory Data Analysis.
- Software & Deployment:** RESTful APIs, Git, Docker, Microsoft Azure, Cloud GPU Infrastructure, Full-Stack Integration, Next.js, React, TypeScript, and Interactive Dashboards.
- Robotics, IoT & Automation Foundations:** Foundational knowledge of robotics concepts, Transformation Matrices, Coordinate Transformations, Denavit–Hartenberg (DH) Parameters, IoT concepts, sensors, connected devices, and intelligent automation.
- Analytical & Problem-Solving Skills:** Analytical Thinking, Problem Solving, Prototyping, Debugging, Experimentation, Technical Documentation, Adaptability, and Cross-functional Collaboration.

WORK EXPERIENCE

Intern Developer (Co-op/Internship) | ThaiDotRun (merged into ASICS), Bangkok

Sept 2025 – Apr 2026

Automated Runner Highlight Extraction System

- Evaluated multiple object detection architectures (including YOLO and RT-DETR), selecting RT-DETR for its Detection Transformer architecture's superior global context modeling in crowded race scenes, paired with BoTSORT for robust subject tracking.
- Implemented a custom spatial weighting algorithm prioritizing camera-center subjects and integrated OCR bib number recognition to dynamically replace track IDs for frame-accurate highlight extraction.
- Achieved an average success rate of approximately 60% in automated highlight video generation, establishing a solid baseline for initial production deployment and future pipeline optimization.

Robustness Shoe Detection & Classification Pipeline

- Designed and developed an end-to-end deep learning pipeline to detect and classify shoe brands from complex race images and video footage.
- Applied unsupervised learning for dataset deduplication to eliminate redundant training frames, using Grounding DINO to localize runners and extract shoe crops for ensemble model training.
- Integrated depth-mapping and keypoint detection to assess image quality and enhance overall prediction reliability.
- Successfully deployed an ensemble classification framework that achieved strong performance on major brands (Adidas, Nike, ASICS), while establishing automated quality control and data re-collection workflows to improve recognition for long-tail brands.

Data Scientist | Lief Capital Asset Management, Bangkok

Short Term: May 2026

- Analyzed and refactored an existing financial data and machine learning codebase on Azure ML, restructuring monolithic workflows across data gathering, preprocessing, feature engineering, model training, and evaluation to improve maintainability and scalable experimentation.
- Optimized the end-to-end data processing and EODHD ingestion pipeline, reducing runtime from over 1 hour to under 30 minutes by adopting Polars and scaling parallel network workers.
- Worked on an XGBoost-based stock ranking pipeline, including sector-relative feature engineering, NDCG performance evaluation, hyperparameter tuning with Optuna, and quarterly backtesting.

PROJECTS

Pivotingual - Multilingual Pivot Neural Machine Translation

July 2026 – Present

- Built a multilingual Neural Machine Translation (NMT) system from scratch using PyTorch and a Pre-LN Transformer Encoder–Decoder architecture for sequence-to-sequence learning.
- Implemented English-centric pivot translation, shared multilingual representations, SentencePiece tokenization, and target-language conditioning for bidirectional translation across multiple languages.
- Applied research-driven Transformer architecture optimization, replacing Post-LN with Pre-Layer Normalization and reducing attention and FFN dimensions to improve training stability and computational efficiency.
- Developed an end-to-end deep learning training pipeline with Mixed Precision (AMP), Multi-GPU training, validation monitoring, checkpointing, and resume-from-checkpoint support.
- Achieved a preliminary validation Perplexity (PPL) of 42.76 in early experiments; model training and downstream translation evaluation are ongoing, with an interactive web demo deployed at sitedotmoss.vercel.app/pivot-translator.

Madonna Discography Analysis

March 2026 – August 2026

- Built an end-to-end unsupervised machine learning and data analytics pipeline using Spotify audio-feature data, including exploratory data analysis (EDA), data preprocessing, deduplication, feature engineering, and correlation analysis.
- Applied Yeo-Johnson power transformation for feature normalization and distribution correction, preventing outlier tracks from forming non-meaningful micro-clusters.
- Benchmarked KMeans clustering and Gaussian Mixture Models (GMM) at K=4 using Silhouette Score and Davies-Bouldin Index, and compared GMM covariance structures using BIC/AIC model selection.
- Designed a statistical validation framework using bootstrap resampling, 50-seed stability analysis, feature-subset sensitivity testing, and per-sample silhouette distributions to evaluate clustering robustness and reproducibility.
- Applied UMAP dimensionality reduction with topology-sensitivity analysis and automatically generated interpretable cluster labels from per-cluster z-scores.

Sitedotmoss ([site\(.\).moss](https://site(.).moss))

December 2025 – Present

- Developed and deployed a full-stack web application using Astro (SSR), React 19, TypeScript, and Turso/LibSQL, featuring a responsive dashboard, blog CRUD system, and integrated AI/ML inference demos.
- Designed a normalized SQLite relational database schema with 5 tables and composite indexes using Drizzle ORM, implementing 8 RESTful API endpoints for full CRUD operations.
- Implemented authentication and session management using PBKDF2 password hashing and secure session handling.
- Integrated real-time YOLOv12 object detection inference from the Capstone Project through Hugging Face Spaces and Gradio REST API, enabling AI model deployment within the web application.
- Engineered modular React components with optimistic UI updates and built an interactive data visualization dashboard using Plotly.js, alongside custom CSS/canvas animations.

Capstone Project (Thesis) — Detection of White Blood Cells in Raptors

August 2024 – August 2025

- Developed deep learning and computer vision object detection models for automated detection of six blood-cell classes in raptor blood smear images using 1,180 manually annotated images.
- Benchmarked YOLO-11l, YOLO-12l, YOLO-26l, and RTDETR-l across five data augmentation strategies, evaluating model performance using mAP, precision, and recall. Achieved a best baseline performance of $\text{mAP}@0.5 = 0.957$ and $\text{mAP}@0.5:0.95 = 0.746$, with 93.4% recall, using YOLO-26l.
- Evaluated Mosaic, MixUp, CLAHE, photometric, and geometric data augmentation techniques and applied Grad-CAM for Explainable AI (XAI) and model interpretability.
- Built and deployed a full-stack AI inference application using Next.js, React, and PostgreSQL (Supabase) for batch image uploads and automated object detection results.
- Manuscript, “YOLO-Based Detection in Wildlife Hematology: Enhancing White Blood Cell Recognition with Data Augmentation,” submitted to *Computer Methods and Programs in Biomedicine* (CMPB-D-26-04827).